

Part A

(i) Name different types of transmission media.

Transmission media are mainly of two types:

1. **Guided (Wired) Media** – Twisted pair cable, Coaxial cable, Optical fiber.
2. **Unguided (Wireless) Media** – Radio waves, Microwaves, Infrared, Satellite.

(ii) What is transmission impairment in data communication?

(iii) Define Parity Check.

(iv) Define Checksum.

(v) Distinguish between detection and correction.

Error Detection

Finds whether an error has occurred.

Does not correct the damaged data.

Example: Parity check, Checksum.

Error Correction

Finds and fixes the error.

Corrects the damaged data.

Example: Hamming Code, Reed-Solomon Code.

(vi) What do you understand by routing?

Routing is the process of selecting the **best path** for transferring data packets from the source to the destination through a network.

- It is mainly performed by **routers**.
- It uses routing information to select an appropriate path.
- It helps deliver packets efficiently.

(vii) Distinguish between unicast and multicast.

Unicast

One sender sends data to one receiver.

One-to-one communication.

Example: Sending an email to one person.

Multicast

One sender sends data to a group of receivers.

One-to-many communication.

Example: Live video sent to a group of users.

(viii) Explain Quality of Service (QoS) for transport layer.

Quality of Service (QoS) refers to the ability of the transport layer to provide reliable and efficient data transmission.

It considers:

Bandwidth – Amount of data transmitted.

Delay – Time taken to deliver data.

Reliability – Proper and error-free delivery.

Jitter – Variation in packet delay.

(ix) Briefly discuss any two transport layer services.

Two important transport layer services are:

1. Reliable Data Transfer:

It ensures that data reaches the destination correctly and in the proper order.

2. Flow Control:

It controls the rate of data transmission so that a fast sender does not overwhelm a slow receiver.

(x) What is the role of cookies in World Wide Web?

Cookies are small pieces of data stored by a website in the user's browser.

They are used to:

- Remember **login/session information**.
- Store user **preferences**.
- Track user activity and provide personalized services.

(xi) What is E-mail privacy?

E-mail privacy means protecting e-mail messages and user information from unauthorized access.

It involves:

- **Authentication** of users.
- **Encryption** of messages.
- Protection against unauthorized reading or modification of e-mails.

(i) List all the layers of TCP/IP Model.

(ii) Define Network Topology and list its types.

(iii) Why UDP is used instead of TCP for video/audio transmission?

UDP is preferred for video/audio because:

- It is **faster** and has low overhead.
- It does not require **connection establishment**.
- It does not retransmit lost packets, so there is **less delay**.
- Real-time applications prefer continuous transmission over perfect delivery.

Example: Live video streaming and online gaming.

(iv) Define Framing and the reason for its need.

Framing is the process of dividing a continuous stream of bits into manageable units called **frames** at the Data Link Layer.

Need for framing:

- Identifies the **start and end** of data.
- Helps in **error detection**.
- Makes data transmission more organized and reliable.

(v) List out the name of all random access protocols.

The main random access protocols are:

1. ALOHA
2. Slotted ALOHA
3. CSMA
4. CSMA/CD
5. CSMA/CA

(vi) What is FTP?

FTP (File Transfer Protocol) is an application-layer protocol used to **transfer files** between a client and a server over a network.

It can be used to:

- Upload files to a server.
- Download files from a server.

- Manage files and directories on a remote server.

(vii) Define Socket Address.

A **socket address** is the combination of an **IP address** and a **port number** used to uniquely identify a communication endpoint.

Example: 192.168.1.10 : 8080

Here, 192.168.1.10 is the IP address and 8080 is the port number.

(viii) Which protocols does data link layer follow for collision avoidance and detection?

The Data Link Layer uses:

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) – used for **collision detection**.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) – used for **collision avoidance**, especially in wireless networks.

(ix) Define ARP.

ARP (Address Resolution Protocol) is used to find the MAC address corresponding to an IP address in a local network.

- Example: If a device knows the IP address 192.168.1.5, ARP helps find the MAC address of that device.

(x) What is Line Coding?

Line coding is the process of converting digital data (bits) into **digital signals** suitable for transmission over a communication medium.

Examples: NRZ, Manchester, and Differential Manchester coding.

(i) What is network hardware? Provide examples.

Network hardware refers to the physical devices used to connect computers and other devices and to transmit data over a network.

Examples:

Router

Switch

Hub

Modem

Network Interface Card (NIC)

(ii) What are the differences between Star and Ring topology?

Star Topology

All devices are connected to a central hub/switch.

Failure of one cable usually affects only one device.

Easy to install and manage.

Ring Topology

Each device is connected to two other devices, forming a ring.

Failure of one device/link can affect the entire network.

More difficult to troubleshoot.

(iii) Define Analog Signals.

(iv) Explain Digital Modulation.

(v) What is error detection in data communication?

(vi) What is a checksum?

A **checksum** is an error-detection value calculated by adding the data blocks together.

The sender transmits the checksum along with the data. The receiver recalculates it and compares the values to **detect errors**.

(vii) Define piggybacking in data communication.

(viii) Explain congestion control in networking.

Congestion control is the process of controlling the amount of data entering a network to prevent excessive traffic.

It helps to:

- Avoid **packet loss**.
- Reduce **network delay**.
- Maintain efficient network performance.

(ix) Define User Datagram Protocol (UDP).

UDP (User Datagram Protocol) is a **connectionless and unreliable transport-layer protocol**.

- It does not guarantee delivery or ordering of packets.
- It has low overhead and is fast.

Examples: Video streaming, online gaming and DNS.

(x) What is the World Wide Web (WWW)?

The **World Wide Web (WWW)** is a system of interconnected web pages and resources available through the Internet.

- Web pages are accessed using **web browsers**.
- It uses protocols such as **HTTP/HTTPS**.
- Web resources are identified using **URLs**.

Unit 1

Q What is the difference between OSI Model and TCP/IP model?

OSI Model

1. OSI stands for Open Systems Interconnection.
2. It has 7 layers.
3. Layers are: Application, Presentation, Session, Transport, Network, Data Link and Physical.
4. It was developed by ISO.
5. It is mainly a reference model.
6. Session and Presentation are separate layers.

TCP/IP Model

TCP/IP stands for Transmission Control Protocol/Internet Protocol.

It has 4 layers.

Layers are: Application, Transport, Internet and Network Access.

It was developed by DARPA.

It is a protocol suite as well as a model.

Session and Presentation functions are included in the Application layer.

OSI Model

- 7. Physical and Data Link are separate layers.
- 8. It is protocol-independent.
- 9. It is mainly used for learning, design and understanding networks.
- 10. It provides a more detailed and structured approach to networking.

TCP/IP Model

- They are combined into the Network Access layer.
- It is based on protocols such as TCP, IP, UDP, HTTP, etc.
- It is widely used in practical Internet communication.
- It provides a simpler and more practical approach to networking.

Q Explain the synchronization problem. How to solve it? Explain with suitable diagrams.

Synchronization Problem

Synchronization means maintaining proper coordination between the sender and receiver during data transmission.

The main problem is that the receiver receives a continuous stream of bits and must know **where each bit, character, or frame starts and ends**. If the receiver loses synchronization, it may read the data incorrectly.

Types of Synchronization Problems

Bit Synchronization:

The receiver must identify the beginning and ending of each bit.

Character Synchronization:

The receiver must identify the beginning and ending of each character or byte.

Frame Synchronization:

The receiver must identify the beginning and ending of each complete frame.

Methods to Solve Synchronization Problem

1. Start and Stop Bits:

In asynchronous transmission, a **start bit** is added before the data and a **stop bit** after the data. The receiver uses these bits to identify the beginning and end of a character.

2. Synchronization Bits/Preamble:

In synchronous transmission, a special **synchronization pattern or preamble** is sent before the actual data. It helps the receiver synchronize its clock with the sender.

3. Line Coding:

Special encoding methods such as **Manchester encoding** provide signal transitions that help the receiver identify bit boundaries and maintain synchronization.

4. Clock Synchronization:

The sender and receiver use properly synchronized clocks so that the receiver samples the incoming signal at the correct time.

Q List major functions performed at each layer of TCP/IP model. What are the major differences in the services at data link layer and transport layer?

- Major Functions of Each Layer of TCP/IP Model(In notes)
- Difference between Data Link Layer and Transport Layer Services

Data Link Layer

Provides **node-to-node (hop-to-hop)** delivery.

Works between two directly connected devices.

Uses **MAC addresses**.

Performs framing and error detection.

Example: Ethernet, Wi-Fi.

Transport Layer

Provides **end-to-end** delivery.

Works between source and destination applications.

Uses **port numbers**.

Performs segmentation, flow control and error/reliability control.

Example: TCP, UDP.

Q Discuss different wireless propagation modes with neat sketches. Mention the range of frequencies for each mode.

Wireless propagation is the way in which radio signals travel from a transmitting antenna to a receiving antenna.

There are mainly **three propagation modes**:

1. Ground Wave Propagation

In ground wave propagation, radio waves travel **along the surface of the Earth**.

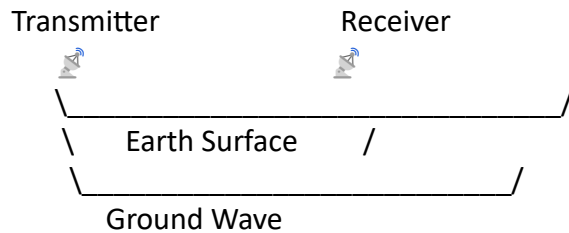
It follows the curvature of the Earth.

It is suitable for **long-distance communication** at lower frequencies.

It is mainly used for **AM broadcasting and navigation**.

Approximate frequency range: **3 kHz – 3 MHz**.

Sketch:



2. Sky Wave Propagation

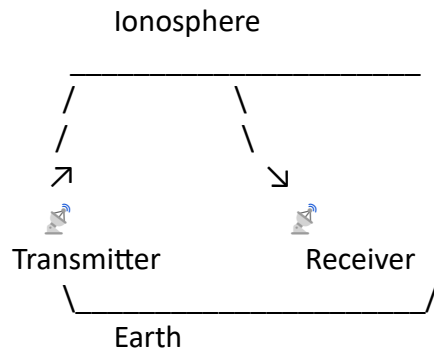
In sky wave propagation, radio waves are sent upward and are **reflected/refracted by the ionosphere** back toward Earth.

It can cover **very long distances**.

It is mainly used for **short-wave radio and international broadcasting**.

Approximate frequency range: **3 MHz – 30 MHz**.

Sketch:



3. Line-of-Sight (Space Wave) Propagation

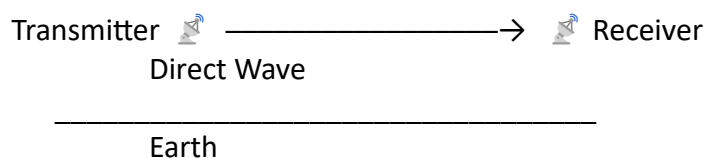
In this mode, radio waves travel **directly from the transmitting antenna to the receiving antenna** without reflection from the ionosphere.

It requires the antennas to be within **line of sight**.

It is used for **TV broadcasting, satellite communication, radar and mobile communication**.

Approximate frequency range: **30 MHz – 300 GHz**.

Sketch:



Summary

Propagation Mode	Path	Approx. Frequency Range
Ground Wave	Along Earth's surface	3 kHz – 3 MHz
Sky Wave	Via ionosphere	3 MHz – 30 MHz

Propagation Mode	Path	Approx. Frequency Range
Line-of-Sight / Space Wave	Direct path	30 MHz – 300 GHz

- Q Explain Network Topologies in Detail with Advantages and Disadvantages
- Q Explain Analog to Digital Conversion and Digital to Digital Conversion with Various Methods.

A. Analog to Digital Conversion

Analog to Digital Conversion (ADC) is the process of converting a continuous **analog signal into digital data (0s and 1s)** so that it can be processed and transmitted by digital systems.

The main steps are:

Sampling → Quantization → Encoding

1. Sampling

Sampling means taking measurements of the analog signal at regular time intervals.

According to the **Nyquist theorem**, the sampling frequency should be at least **twice the highest frequency** of the analog signal.

2. Quantization

The sampled values are converted into a limited number of discrete amplitude levels.

3. Encoding

Each quantized value is represented by a **binary code** such as 0s and 1s.

Methods of Analog to Digital Conversion

1. Pulse Code Modulation (PCM):

In PCM, the analog signal is **sampled, quantized and encoded** into binary data. It is widely used in digital audio and telephone systems.

2. Delta Modulation (DM):

Delta modulation represents the **difference between the current sample and the previous sample** using a single bit, usually 0 or 1.

3. Adaptive Delta Modulation (ADM):

ADM is an improved form of delta modulation in which the **step size is adjusted according to the changes in the input signal**.

B. Digital to Digital Conversion

Digital to Digital Conversion is the process of converting digital data into a **digital signal** suitable for transmission through a communication medium.

The major techniques are:

1. Line Coding

Line coding converts binary data into digital signal patterns.

Common line coding methods include:

- **Unipolar:** Uses one voltage level to represent a bit.
- **Polar:** Uses positive and negative voltage levels.
- **Bipolar:** Uses three voltage levels, such as positive, zero and negative.
- **Manchester:** Each bit has a transition in the middle, which also helps synchronization.
- **Differential Manchester:** Uses transitions to represent bits and provides synchronization.

2. Block Coding

In block coding, a group of **m data bits is converted into n coded bits ($n > m$)**.

It adds extra bits to provide:

- Better synchronization.
- Error detection.
- More reliable transmission.

Example: 4B/5B coding converts every 4 bits into 5 bits.

3. Scrambling

Scrambling replaces long sequences of unwanted voltage levels with special patterns to maintain **synchronization** without increasing the number of bits significantly.

Examples:

- B8ZS
- HDB3

Q **What is Pulse Code Modulation (PCM)? Explain all the steps involved in it.**

Pulse Code Modulation (PCM) is a technique used to convert an **analog signal into digital data**.

It is widely used in digital telephone systems, audio recording and digital communication.

The PCM process consists of **three main steps: Sampling, Quantization and Encoding**.

1. Sampling

Sampling is the process of measuring the amplitude of an analog signal at **regular time intervals**.

The analog signal is converted into a sequence of samples.

According to the **Nyquist theorem**, the sampling frequency should be at least **twice the highest frequency** of the signal.

If the highest frequency is f_m , then:

Sampling frequency $\geq 2f_m$

2. Quantization

Quantization is the process of converting the sampled amplitudes into a **finite number of discrete levels**.

Each sample is assigned to the nearest available level.

Due to this approximation, a small **quantization error** may occur.

More quantization levels provide better quality.

3. Encoding

Encoding is the process of converting each quantized level into a **binary code** consisting of 0s and 1s.

For example, if there are 8 quantization levels, each level requires:

$\log_2 8 = 3$ bits

Thus, the analog signal is finally represented as a stream of digital bits.

PCM Process

Analog Signal → Sampling → Quantization → Encoding → Digital Data

Advantages of PCM

High noise immunity.

Good quality and reliability.

Digital signals can be easily stored and processed.

Suitable for long-distance communication.

Q **How is digital data converted into digital signals? Explain all the line coding schemes in detail.**

Digital-to-Digital conversion is the process of converting **digital data (0s and 1s) into digital signals** so that it can be transmitted through a communication medium.

This process is called **Line Coding**.

Main Line Coding Schemes

There are **3 main types**:

1. Unipolar Coding

In unipolar coding, **only one voltage level and zero voltage** are used.

Unipolar NRZ

1 → Positive voltage

0 → Zero voltage

Signal does not return to zero during the bit time.

Easy to remember:

👉 **1 = +V, 0 = 0**

Disadvantage: Poor synchronization and has a DC component.

2. Polar Coding

Polar coding uses **two voltage levels: positive (+V) and negative (-V)**.

(a) NRZ-L

One voltage represents **1**.

The other voltage represents **0**.

Signal remains at the same level during the bit.

👉 **L = Level**

(b) NRZ-I

Data is represented by a **change or no change** in voltage.

A transition occurs for one binary value.

No transition represents the other value.

👉 **I = Invert**

(c) RZ (Return to Zero)

Signal represents the bit and then **returns to zero** before the bit time ends.

Gives better synchronization.

Requires more bandwidth.

👉 **RZ = Return to Zero**

(d) Manchester

There is always a **transition in the middle of every bit**.

The direction of transition represents 0 or 1.

Provides good synchronization.

👉 **Manchester = Middle transition**

(e) Differential Manchester

There is always a transition in the **middle of each bit**.

Data is represented by whether there is a **transition at the beginning** of the bit or not.

Provides good synchronization.

3. Bipolar Coding

Bipolar coding uses **three voltage levels: +V, 0 and -V**.

(a) AMI (Alternate Mark Inversion)

0 → Zero voltage

1 → Alternately +V and -V

Example:

1 → +V, 1 → -V, 1 → +V

👉 **AMI = 1s keep changing + and -**

(b) Pseudoternary

1 → Zero voltage

0 → Alternately +V and -V

👉 It is basically **opposite of AMI**

Q **Define Digital Modulation. Explain Different Types of Digital Modulation Techniques — 8 Marks**

Definition

The three main types are:

1. **ASK – Amplitude Shift Keying**
2. **FSK – Frequency Shift Keying**
3. **PSK – Phase Shift Keying**
(all graph and definition in notes)

Comparison

ASK	FSK	PSK
Changes amplitude	Changes frequency	Changes phase
Simple technique	More resistant to noise than ASK	More reliable than ASK
More affected by noise	Requires more bandwidth	Requires less bandwidth than FSK
Used in low-speed communication	Used in modems and wireless systems	Used in modern digital communication

- Q **Compare and contrast the various network topologies such as bus, star, and ring.**
- Q **Describe the seven layers of the OSI model and their functions.**
- Q **Define digital modulation. Explain different types of digital modulation techniques.**
- Q **What are periodic and non-periodic signals in networking? Explain which type of analog and digital signals are preferred and give their reason.**

Unit 2

Q **Explain linear codes and how errors can be detected and corrected using linear code techniques**

A **linear code** is an error-control coding technique in which data bits are combined with extra **redundant bits** to form a codeword.

The important property is that the **sum (XOR) of any two valid codewords is also a valid codeword.**

How Linear Codes Work

Suppose the original data is:

Data → Encoder → Codeword → Transmission → Decoder → Data

The encoder adds redundant bits to the original data. These extra bits help the receiver **detect and correct errors.**

Error Detection

At the receiver, the received codeword is checked using a **parity-check matrix (H).**

The receiver calculates the **syndrome**:

$$S = R \times H^T$$

where:

R = received codeword

H = parity-check matrix

S = syndrome

Cases:

S = 0: No error is detected.

$S \neq 0$: An error is detected.

Thus, the syndrome tells the receiver whether the received data is correct or not.

Error Correction

If an error is detected, the syndrome can be used to **find the position of the error**.

For example, in a simple linear block code:

- Receiver receives the codeword.
- It calculates the syndrome.
- If the syndrome is zero $\rightarrow$ data is correct.
- If the syndrome is non-zero $\rightarrow$ error has occurred.
- The syndrome is compared with known error patterns.
- The position of the error is identified.
- The incorrect bit is **flipped ($0 \rightarrow 1$ or $1 \rightarrow 0$)**.
- The original data is obtained.

Q **What is the sliding window protocol? What should be the size of the window? Explain.**

Sliding window(in notes).

The window size depends mainly on the **sequence number field** and the type of sliding-window protocol.

For Go-Back-N

If there are **m bits** available for sequence numbers:

Maximum Window Size = $2^m - 1$

For Selective Repeat

For Selective Repeat, the window size should be at most:

Window Size $\leq 2^{m-1}$

- Q **Explain Pure ALOHA and Slotted ALOHA. Show that Slotted ALOHA has a maximum throughput of twice the Pure ALOHA maximum throughput.**
- Q **Explain the working of Stop-and-Wait protocol with the help of suitable diagrams.**
- Q **Explain Sliding Window Protocol in detail.**
- Q **Explain Pure ALOHA and Slotted ALOHA in detail with their differences.**
- Q **What is the need of flow control? Explain Stop-and-Wait ARQ and Sliding Window Protocol.**

1. Need of Flow Control

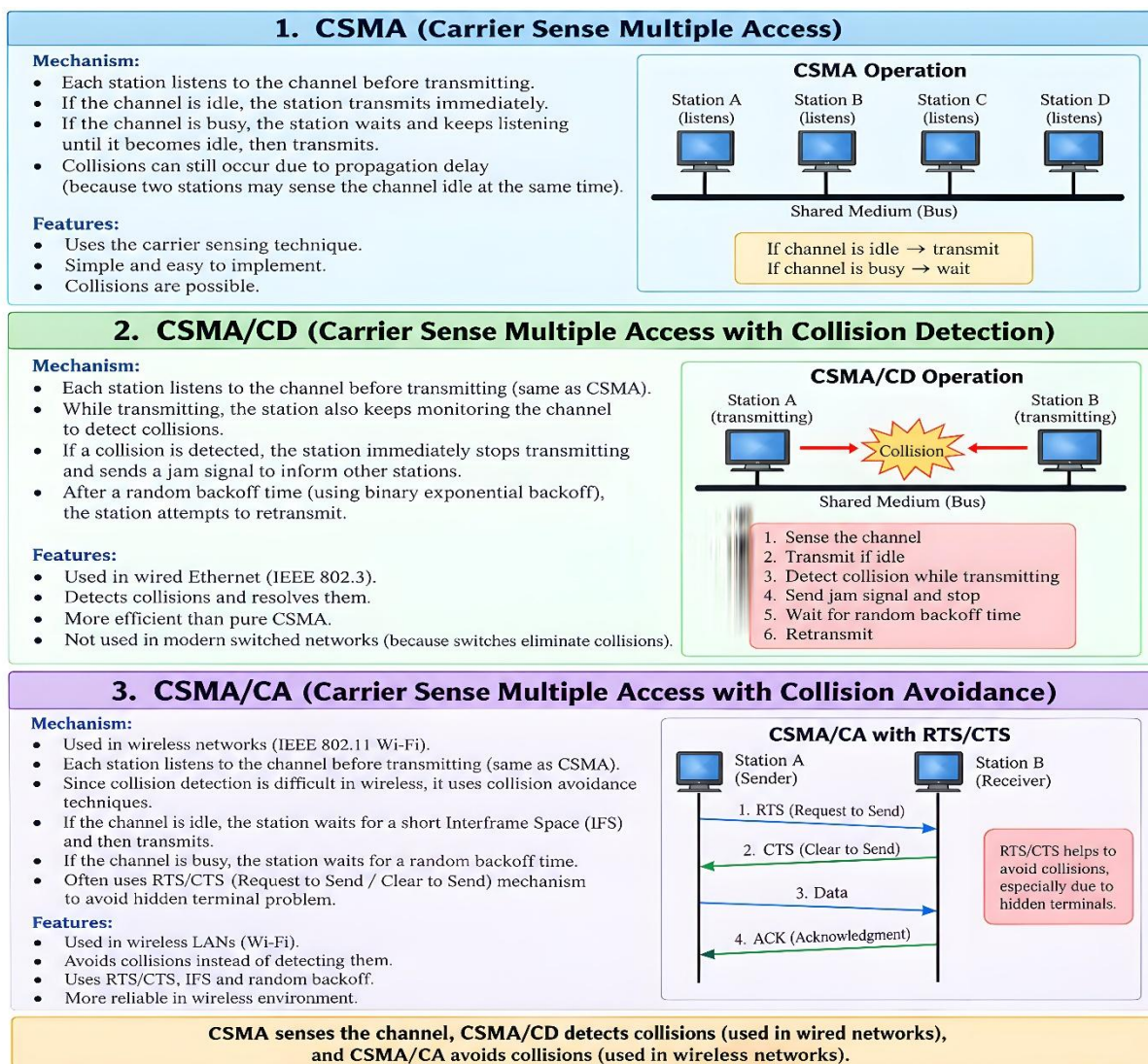
Flow control is a technique used to control the rate at which the sender sends data to the receiver.

It is needed because the **sender and receiver may have different processing speeds**. If a fast sender sends data faster than the receiver can process, the receiver's buffer may become full and data may be lost.

Need for flow control:

1. Prevents **receiver buffer overflow**.
2. Prevents **loss of data**.
3. Controls the speed of data transmission.
4. Maintains proper communication between sender and receiver.
5. Improves the **efficiency and reliability** of data transfer.

Q Elaborate the mechanism of CSMA, CSMA/CD and CSMA/CA with diagram.



Q Describe the Stop-and-Wait ARQ Protocol. How does it ensure reliable data transmission?

How does it ensure reliable transmission?

Method	Purpose
ACK	Confirms that the frame was received correctly.
Timeout	Detects a lost frame or lost ACK.
Retransmission	Resends a frame when an error or loss occurs.
Sequence Number	Identifies frames and prevents duplicate delivery.
Error Detection	Detects damaged frames using techniques such as CRC.

Q Explain the Checksum. Suppose that a message 1001 100 10100011 is transmitted using Internet Checksum with 4-bit words. What is the value of the checksum?

Checksum is an error-detection technique used in data communication. The sender divides the data into equal-sized words, adds them using **one's complement addition**, and then takes the **one's complement of the sum**. This value is called the **checksum**.

The receiver performs the same calculation to check whether an error has occurred.

Given

Message:

1001 100 10100011

Since the word size is **4 bits**, first divide the message into 4-bit words:

1001 | 1001 | 0100 | 0011

Step 1: Add the 4-bit words

```
  1001
+ 1001
-----
 10010
```

There is a carry 1. Add it back to the lower 4 bits:

```
  0010
+    1
-----
  0011
```

Now add the next word:

```
  0011
+ 0100
-----
  0111
```

Add the last word:

```
  0111
+ 0011
-----
 1010
```

So, the final sum is:

1010

Step 2: Take 1's Complement

Change **0 → 1** and **1 → 0**:

Sum = 1010

1's Comp. = 0101

✅ Final Answer

Checksum = 0101

Q Consider the delay of Pure ALOHA versus Slotted ALOHA at low load. Which one is less? Explain your answer.

Pure ALOHA vs Slotted ALOHA: Delay at Low Load

At low load, Pure ALOHA has less delay than Slotted ALOHA.

Reason:

Pure ALOHA

A station can transmit **at any time** when it has data.

It does not have to wait for the next slot.

At low load, collisions are very rare.

Therefore, the waiting time is very small.

Lower delay at low load.

Simple Example

Slotted ALOHA

A station can transmit **only at the beginning of a time slot**.

It may have to wait for the next slot to begin.

At low load, collisions are also very rare.

There is an additional waiting time for the next slot.

Higher delay at low load.

Suppose a station gets a packet **in the middle of a slot**:

- **Pure ALOHA**: It can **transmit immediately**.
- **Slotted ALOHA**: It must **wait until the next slot begins**.

Therefore:

At low traffic/load, Pure ALOHA has lower delay because it allows immediate transmission, while Slotted ALOHA requires waiting for the next slot.

- ❖ **What are the different types of errors that can occur during data transmission? (in notes).**

Difference

Single-Bit Error

Only **one bit** is changed.

Less common in many communication systems.

Easy to detect and correct.

Example: 1010 → 1000

Burst Error

Two or more bits are affected.

More common in data communication.

Usually requires stronger error-detection methods.

Example: 10110110 → 10001010

Unit 3

- ❖ **Explain the difference between IPv4 and IPv6 with suitable diagrams.**

Difference between IPv4 and IPv6 — 8 Marks

IPv4 (Internet Protocol version 4) and **IPv6 (Internet Protocol version 6)** are protocols used to identify devices and deliver packets over a network.

Suitable Diagrams

IPv4 Address: 32 bits

32 bits

32 bits			
8 bit	8 bit	8 bit	8 bit
192	168	1	10

Example: 192.168.1.10

IPv6 Address: 128 bits

128 bits

16 bit	16 bit	16 bit	16 bit	16 bit	16 bit	16 bit	16 bit
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Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv4 vs IPv6

IPv4

1. IPv4 uses a **32-bit** address.
2. Example: **192.168.1.10**
3. Provides about **4.3 billion addresses**.
4. Address is written in **decimal** format.
5. Header size is **20–60 bytes**.

IPv6

- IPv6 uses a **128-bit** address.
- Example: **2001:db8::1**
- Provides a very large number of addresses, about **3.4×10^{38}** .
- Address is written in **hexadecimal** format.
- Basic header size is **40 bytes**.

IPv4

6. Supports **broadcast** communication.
7. **ARP** is used to find the MAC address from an IP address.
8. Fragmentation can be performed by the **sender and routers**.
9. Security support through IPsec is available but not inherently required.
10. Uses **DHCP** for automatic address configuration.
11. Address space is becoming insufficient due to the large number of devices.

IPv6

Does **not use broadcast**; it uses multicast and anycast.

Neighbor Discovery Protocol (NDP) performs this function.

Fragmentation is performed only by the **source device**.

IPv6 was designed with stronger integrated support for IPsec.

Supports **SLAAC** and DHCPv6 for address configuration.

Provides a **much larger address space**, solving IPv4 address exhaustion.

Main Advantages of IPv6

- Provides a **huge number of IP addresses**.
- Simplifies the IP header.
- Supports **automatic address configuration**.
- Better support for **multicast and anycast**.
- Designed to support the growing number of Internet-connected devices.

❖ Routing Principle

Routing is the process of selecting the **best path** for delivering data packets from a source device to the destination through a network.

A router examines the available paths and selects the best path based on a **routing metric**, such as:

- Number of hops
- Distance/cost
- Delay
- Bandwidth
- Reliability

Basic Routing Process

1. Packet arrives at a router.
2. Router checks the **destination IP address**.
3. It looks in its **routing table**.
4. It selects the best available path.
5. The packet is forwarded to the **next router**.
6. This continues until the packet reaches the destination.

❖ . Distance Vector Routing Algorithm

Distance Vector Routing (DVR) is a dynamic routing algorithm in which each router maintains information about the **distance (cost) to each destination and the next router (next hop)**.

It is called **Distance Vector** because:

- **Distance** = cost to reach the destination.
- **Vector** = direction/next hop to reach the destination.

❖ **Routing protocols** are used by routers to find and maintain the **best path** for sending packets from source to destination.

There are three important routing methods:

1. Distance Vector Routing

In **Distance Vector Routing**, each router maintains a table containing the **distance (cost)** to each destination and the **next hop**.

Working:

- Each router knows the cost of its directly connected networks.
- Routers exchange their routing information with their **neighboring routers**.
- Each router calculates the best path using the **minimum cost**.
- The routing table is updated whenever a better path is found.

Example: RIP (Routing Information Protocol)

Easy idea:

👉 Router asks its neighbors: "How far is the destination from you?"

Advantages:

- Simple to implement.
- Requires less processing.

Disadvantages:

- Slow convergence.
 - Can suffer from **count-to-infinity** problem.
-

2. Link State Routing

In **Link State Routing**, every router builds a complete map of the network by learning about the **state and cost of all links**.

Working:

- Each router discovers its neighboring routers.
- It determines the cost of each link.
- It shares this information with all routers in the network.
- Each router builds a complete network map.
- It uses **Dijkstra's algorithm** to find the shortest/best path.

Example: OSPF (Open Shortest Path First)

Easy idea:

👉 Every router gets a map of the whole network and finds the best route.

Advantages:

- Faster convergence.
- More accurate routing.
- Less chance of routing loops.

Disadvantages:

- More memory and processing are required.
 - More complex than distance vector routing.
-

3. Path Vector Routing

In **Path Vector Routing**, routers maintain information about the **complete path** to reach a destination rather than only the distance.

Working:

- A router advertises a destination along with the **path of networks/routers** to reach it.
- Other routers receive this path information.
- The router selects a suitable path based on policies and path information.
- If a router detects its own identity in a path, it rejects that route to avoid a **routing loop**.

Example: BGP (Border Gateway Protocol)

Easy idea:

👉 Router knows the complete path: $A \rightarrow B \rightarrow C \rightarrow D$.

Advantages:

- Prevents routing loops using path information.
- Suitable for **large networks and the Internet**.

- Supports routing policies.

Disadvantages:

- More complex.
- Routing decisions can take more time.

Difference Between the Three

Distance Vector	Link State	Path Vector
Knows distance and next hop .	Knows the complete network topology .	Knows the complete path to destination.
Shares information with neighbors .	Shares link-state information throughout the network.	Shares path information with other routers.
Uses distance/cost to select route.	Uses Dijkstra's algorithm .	Uses path and routing policies.
Example: RIP	Example: OSPF	Example: BGP
Suitable for smaller networks.	Suitable for large internal networks.	Mainly used between different autonomous systems.

Difference between ARQ and RARQ

ARQ	RARQ
ARQ stands for Automatic Repeat Request.	RARQ stands for Relay-Assisted Automatic Repeat Request.
If data has an error, the sender sends it again .	If data has an error, a relay helps to send it again .
Communication is mainly between sender and receiver .	Communication involves sender, relay and receiver .
It does not require a relay.	It requires a relay to help in transmission.
It is simple to use.	It is more complex than ARQ.
Mainly used for normal data communication.	Mainly useful in wireless/multi-hop networks .

❖ Shortest Path Routing Algorithm — 8 Marks

Definition

Shortest Path Routing is a routing technique in which the router selects the **best/shortest path** from the source to the destination based on the lowest cost.

The cost may be based on **distance, number of hops, delay, bandwidth**, etc.

Working

The shortest path algorithm generally works as follows:

1. Start from the **source node**.
2. Assign distance **0** to the source and infinity (∞) to other nodes.
3. Select the node having the **smallest known distance**.
4. Calculate the distance to its neighboring nodes.
5. If a shorter path is found, **update the distance**.
6. Repeat the process until the destination is reached.
7. The resulting path with minimum cost is selected as the **shortest path**.

Dijkstra's algorithm is commonly used for shortest path routing.

Example

Consider the following network:

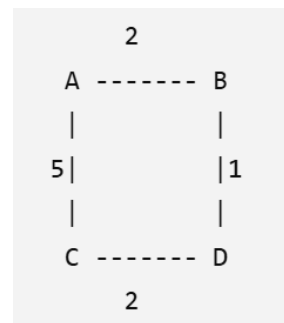
To find the shortest path from **A to D**:

- $A \rightarrow B \rightarrow D = 2 + 1 = 3$
- $A \rightarrow C \rightarrow D = 5 + 2 = 7$

Therefore, the shortest path is:

A → B → D

and its total cost is **3**.



Advantages

- Finds the path with minimum cost.
- Reduces transmission delay.
- Makes efficient use of network resources.
- Suitable for large networks.

❖ Congestion Control and Congestion Prevention Policies — 8 Marks

Congestion occurs when too many packets are present in a network and the network cannot handle them efficiently.

It causes:

- Packet loss
- Increased delay
- Reduced network performance
- Buffer overflow

Congestion Control

Congestion control is the process of controlling the amount of traffic entering a network so that the network does not become overloaded.

Policies for Congestion Prevention

The main congestion prevention policies at the network layer are:

1. Retransmission Policy

Packets should not be retransmitted unnecessarily.

- Unnecessary retransmissions increase network traffic.
- Proper retransmission methods help reduce congestion.

2. Window Policy

The number of packets sent before receiving acknowledgment can be controlled.

- A smaller window reduces network traffic.
- It helps prevent the network from becoming overloaded.

3. Acknowledgment Policy

Acknowledgments should be sent efficiently.

- Sending too many ACKs increases traffic.
- Delayed or combined acknowledgments can reduce unnecessary traffic.

4. Discarding Policy

When congestion becomes severe, routers may **discard some packets**.

- This prevents router buffers from becoming completely full.
- Less important packets may be discarded first.

5. Admission Policy

The network can **refuse new connections or traffic** when it is already heavily loaded.

- This prevents additional traffic from entering a congested network.
- It helps maintain network performance.

6. Routing Policy

Routers can select **alternative paths** when the current path is congested.

- Traffic is distributed among different paths.
- This reduces the load on a particular route.

❖ Link State Routing Algorithm — 8 Marks

Link State Routing is a routing method in which every router obtains information about the **complete network topology** and uses it to find the shortest path.

Example: OSPF (Open Shortest Path First).

Working of Link State Routing

1. **Discover Neighbors:**
Each router identifies its directly connected neighboring routers.
2. **Measure Link Cost:**
The router determines the cost of each link. Cost may depend on bandwidth, delay, etc.
3. **Create Link State Packet:**
Each router creates a packet containing information about its neighbors and link costs.
4. **Distribute Information:**
The link-state information is distributed to all routers in the network.
5. **Build Network Map:**
Every router creates a complete map of the network.
6. **Find Shortest Path:**
Each router uses **Dijkstra's Shortest Path Algorithm** to find the best path.
7. **Create Routing Table:**
The best paths are stored in the routing table.

Simple Example

For A to D:

- $A \rightarrow B \rightarrow D = 2 + 1 = 3$
- $A \rightarrow C \rightarrow D = 5 + 2 = 7$

Therefore, the shortest path is:

$A \rightarrow B \rightarrow D$

Advantages

- Provides **fast convergence**.
- Each router has a complete network map.
- Finds efficient shortest paths.
- Less chance of routing loops.
- Suitable for large networks.

Disadvantages

- Requires more **memory and processing power**.
- More complex than distance vector routing.
- Link-state information requires additional network traffic.
- Initial calculation can be computationally expensive.

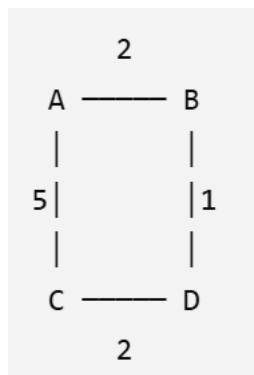
❖ Role of Routing and QoS in Routing — 8 Marks

1. Role of Routing in Computer Networks

Routing is the process of selecting the **best path** for sending data packets from the source to the destination.

Main roles of routing:

1. **Finds the best path** between source and destination.
2. **Forwards packets** from one router to another.
3. Uses routing information such as **distance, cost, delay and bandwidth**.
4. Helps packets reach the **correct destination**.
5. Can select an **alternative path** if the current path fails.
6. Helps in the **efficient use of network resources**.
7. Reduces unnecessary **delay and traffic**.
8. Maintains communication between different networks.



2. Quality of Service (QoS) in Routing

Quality of Service (QoS) refers to the ability of a network to provide **better and controlled performance** for different types of data traffic.

QoS is important for applications such as **video calls, audio calls and live streaming**, where delay and packet loss can affect quality.

Main QoS factors:

1. **Bandwidth:** Amount of data that can be transmitted in a given time.
2. **Delay:** Time taken for a packet to travel from source to destination.
3. **Jitter:** Variation in packet delay.
4. **Packet Loss:** Packets that are lost during transmission.
5. **Reliability:** Ability of the network to deliver data correctly.

Role of QoS in Routing

- Routes important traffic through **better paths**.
- Gives priority to **delay-sensitive traffic** such as voice and video.
- Helps reduce **delay, jitter and packet loss**.
- Provides better and more reliable network performance.

❖ **Leaky Bucket Algorithm**

The **Leaky Bucket Algorithm** is a congestion-control technique used to control the **rate at which packets are sent into the network**.

It works like a bucket with a small hole at the bottom. Packets enter the bucket at different rates, but they leave at a **fixed and controlled rate**.

Advantages

- Controls the flow of packets.
- Reduces congestion.
- Smooths out bursty traffic.
- Prevents sudden overload of the network.
- Simple to implement.

Disadvantage

If packets arrive faster than the fixed output rate for a long time, the **bucket becomes full and packets are lost**.

Unit 4

❖ **TCP: Definition, Features and Three-Way Handshake**

TCP Definition

TCP (Transmission Control Protocol) is a **connection-oriented and reliable transport-layer protocol**. It provides reliable, ordered and error-free delivery of data between two devices.

Features of TCP

1. **Connection-oriented:** A connection is established before data transfer.
2. **Reliable:** Ensures that data reaches the destination correctly.
3. **Error control:** Detects errors and retransmits lost or damaged data.
4. **Flow control:** Controls the speed of data transmission between sender and receiver.
5. **Congestion control:** Controls traffic when the network becomes congested.
6. **Ordered delivery:** Delivers data to the application in the correct order.
7. **Full-duplex:** Both sides can send and receive data at the same time.
8. **Uses acknowledgments:** Receiver sends ACKs to confirm successful data reception.

- The **Three-Way Handshake** is the process used by TCP to **establish a connection** between a client and a server before data transmission.

Steps:

1. SYN – Synchronize

The client sends a **SYN (Synchronize)** segment to the server to request a connection and synchronize the sequence numbers.

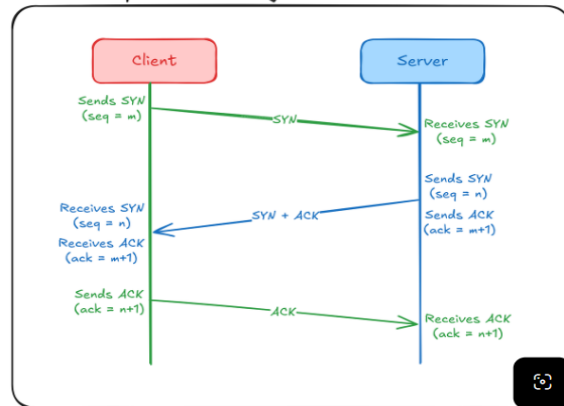
2. SYN + ACK – Synchronize + Acknowledgment

The server sends **SYN + ACK (Synchronize + Acknowledgment)** to accept the connection request and acknowledge the client's SYN.

3. ACK – Acknowledgment

The client sends an **ACK (Acknowledgment)** to confirm the server's response. The TCP connection is now established.

Three-Way Handshaking



❖ UDP Segment, Advantages and Disadvantages over TCP

UDP Definition

UDP (User Datagram Protocol) is a **connectionless and unreliable transport-layer protocol**. It sends data without establishing a connection first.

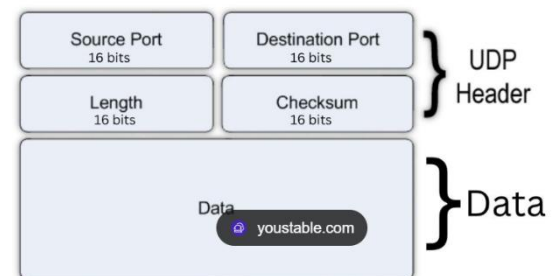
Structure of UDP Segment

The UDP header is very simple and has **4 fields**:

Fields

1. **Source Port**: Identifies the sending application.
2. **Destination Port**: Identifies the receiving application.
3. **Length**: Specifies the total length of the UDP segment.
4. **Checksum**: Used to detect errors in the segment.
5. **Data**: Contains the actual user data.

Structure of User Datagram Protocol Header



Advantages of UDP over TCP

1. **Faster**: UDP has less processing overhead.
2. **No connection establishment**: It does not use a handshake.
3. **Low delay**: Suitable for real-time applications.
4. **Small header**: UDP has only an **8-byte header**.
5. **Suitable for real-time applications**: Used for video/audio streaming, online gaming and DNS.

Disadvantages of UDP over TCP

1. **Not reliable**: It does not guarantee delivery.
2. **No retransmission**: Lost packets are not automatically resent.
3. **No ordering**: Packets may arrive out of order.
4. **No flow control**: It does not control the sender's speed.
5. **No congestion control**: It does not automatically control network congestion.

❖ Flow Control and Buffering as Elements of Transport Protocol

1. Flow Control

Flow control is a mechanism used to control the **speed of data transmission** between the sender and receiver.

The sender may send data faster than the receiver can process it. Flow control prevents the receiver's buffer from becoming full.

Functions of Flow Control:

- Prevents **buffer overflow** at the receiver.
- Controls the sender's transmission speed.
- Prevents **loss of data**.
- Ensures smooth and reliable communication.
- Makes the sender and receiver work at a suitable speed.

Example:

If the sender can send **100 packets/second**, but the receiver can process only **50 packets/second**, flow control slows down the sender to avoid data loss.

2. Buffering

Buffering is the temporary storage of data in a **memory area called a buffer** while it is waiting to be processed or transmitted.

The transport layer uses buffers at both the **sender and receiver sides**.

Types of Buffering:

1. Sender Buffer:

Stores data that has been sent but is **waiting for acknowledgment** or retransmission.

2. Receiver Buffer:

Stores received data temporarily until the receiving application is ready to process it.

Importance of Buffering:

- Handles differences in sender and receiver speeds.
- Stores data temporarily during transmission.
- Helps in **retransmission** of lost data.
- Prevents immediate data loss when data arrives in bursts.

❖ Scenarios for Transport Connection over a Connection-Oriented Network Layer

A **connection-oriented network layer** establishes a logical connection between the source and destination before transferring data.

The transport layer may face different situations while establishing its own connection.

1. Both Connections Are New

The network connection and transport connection are both established normally.

Transport Connection → New

Network Connection → New

The transport layer establishes a new connection using the existing network connection.

2. Network Connection Already Exists

Sometimes the network layer connection is already established.

Transport Connection → New

Network Connection → Already exists

In this case, the transport layer can use the existing network connection instead of creating another one.

3. Network Connection Is Lost

The network connection may fail while the transport connection is still required.

Transport Connection → Existing

Network Connection → Lost

The transport layer must establish a **new network connection** and continue the transport connection if possible.

4. Network Connection Fails and Recovers

The network connection may fail temporarily and later be restored.

The transport layer must determine whether the recovered connection is the **same old connection or a new connection** and should avoid accepting old/duplicate packets.

❖ Multiplexing and De-multiplexing at Transport Layer — 8 Marks

Multiplexing is the process of combining data from **multiple applications** and sending it through the network using the transport layer.

Why is Multiplexing Needed?

- Many applications can work **at the same time** on one computer.
- Each application generates its own data.
- The transport layer collects this data and sends it through the network.
- It makes efficient use of a **single network connection**.
- **Port numbers** are used to identify different applications.

Example:

At the same time, a computer may use a browser, email and online game. The transport layer handles data from all these applications.

Multiplexing

In **multiplexing**, data from different applications is collected by the transport layer and sent to the network.

The transport layer adds **port numbers** to identify the source and destination applications.

De-multiplexing

De-multiplexing is the reverse process of multiplexing.

When data reaches the destination, the transport layer checks the **destination port number** and delivers the data to the correct application.

❖ UDP: Uses, Reasons and UDP Header — 8 Marks

UDP

UDP stands for User Datagram Protocol. It is a **connectionless and unreliable transport-layer protocol**. It sends data without establishing a connection first.

Where is UDP used instead of TCP?

UDP is preferred in applications where **speed and low delay are more important than reliability**.

Examples:

1. **Video streaming** – small packet loss is acceptable, but delay should be low.
2. **Audio/Voice calls** – real-time communication needs fast delivery.
3. **Online gaming** – latest data is more important than retransmitting old data.
4. **DNS (Domain Name System)** – small queries and quick responses are required.
5. **Live broadcasting** – low delay is more important than perfect delivery.

Why UDP is preferred?

- No connection establishment is required.
- Faster than TCP.
- Has a **small 8-byte header**.
- No waiting for acknowledgments.
- Suitable for **real-time applications**.
- Creates less network overhead.

❖ Token Bucket Algorithm

The **Token Bucket Algorithm** controls traffic by generating and storing **tokens** at a fixed rate.

A packet can be transmitted only when a required token is available.

Working

1. Tokens are generated at a **fixed rate**.
2. Tokens are stored in the token bucket.
3. When a packet arrives, it takes the required number of tokens.
4. If tokens are available, the packet is transmitted.
5. If tokens are not available, the packet waits.
6. Stored tokens allow **bursty traffic** for a short time.

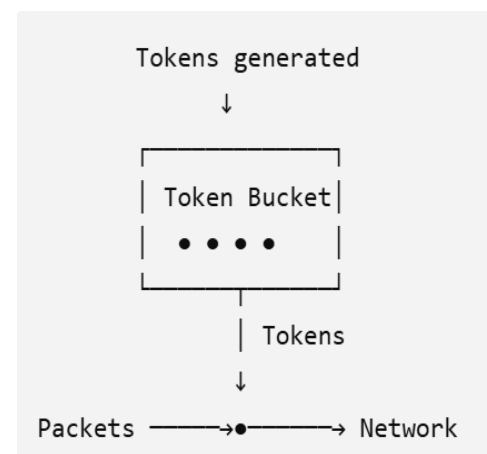
Advantages

- Allows **burst transmission**.
- Controls network traffic.
- Reduces congestion.
- More flexible than Leaky Bucket.

Disadvantage

- More complex than Leaky Bucket.
- Requires token management.

❖ Main Functions of Transport Layer in OSI Model



The **Transport Layer** is the **4th layer of the OSI model**. It provides **end-to-end communication** between applications running on different devices.

Main Functions

1. **Process-to-process delivery:**
Delivers data from one application/process to the correct application on another device.
2. **Segmentation and Reassembly:**
Divides large data into smaller **segments** and joins them again at the destination.
3. **Flow Control:**
Controls the speed of data transmission so that the receiver is not overloaded.
4. **Error Control:**
Detects errors and ensures reliable delivery using **ACK and retransmission**.
5. **Connection Control:**
Establishes, maintains and terminates connections when required.
6. **Multiplexing and De-multiplexing:**
Allows multiple applications to use the network at the same time using **port numbers**.
7. **Congestion Control:**
Controls the amount of data entering the network to reduce congestion.
8. **Reliable Delivery:**
Ensures that data is delivered **correctly and in the proper order** when a reliable service is used.

❖ Connection-Oriented vs Connectionless Communication

Connection-Oriented Communication	Connectionless Communication
A connection is established before data transfer.	No connection is established before data transfer.
Data is sent after setting up a connection.	Data is sent directly.
Provides reliable communication .	Generally provides unreliable communication .
Packets usually arrive in the correct order .	Packets may arrive out of order .
Uses acknowledgments and retransmission.	Usually does not use acknowledgments or retransmission.
Has more overhead.	Has less overhead.
Generally slower due to connection setup.	Generally faster.
Example: TCP (Transmission Control Protocol)	Example: UDP (User Datagram Protocol)

Unit 5

❖ Network Security — 8 Marks

Network Security is the process of protecting a computer network and its data from **unauthorized access, attacks, misuse and damage**.

The main aim of network security is to keep data **safe, private and available** to authorized users.

Principles of Network Security

The main principles are called the **CIA Triad**:

1. Confidentiality

It ensures that data is accessible **only to authorized users**.

Example: Passwords and encryption are used to protect private information.

2. Integrity

It ensures that data is **not changed or modified without permission**.

Example: Error checking and digital signatures help maintain data integrity.

3. Availability

It ensures that network services and data are **available whenever authorized users need them**.

Example: Backup systems and protection against attacks help maintain availability.

Other Important Principles

4. **Authentication:** Verifies the identity of a user or device.
 5. **Authorization:** Determines what an authenticated user is allowed to access.
 6. **Non-repudiation:** Ensures that a person cannot deny an action they performed.
-

Challenges in Implementing Network Security

1. Increasing Cyber Attacks

Viruses, malware, phishing and other attacks are continuously increasing.

2. Unauthorized Access

Attackers may try to access private network resources.

3. Data Theft

Sensitive information such as passwords and financial data may be stolen.

4. Network Complexity

Large networks have many devices, making security difficult to manage.

5. Insider Threats

Authorized users may accidentally or intentionally misuse network resources.

6. Cost

Security tools such as firewalls, antivirus software and monitoring systems can be expensive.

7. New Technologies

Cloud computing, IoT and wireless networks create new security risks.

8. Maintaining Security

Security systems must be regularly **updated and monitored** because new threats appear frequently.

16. (a) DNS — 8 Marks

DNS Definition

DNS (Domain Name System) is a system that converts a **domain name into an IP address**.

For example:

www.google.com → IP address

It is easier for people to remember names than numerical IP addresses.

1. Domain Name

A **domain name** is a human-readable name used to identify a website or host on the Internet.

Example:

www.example.com

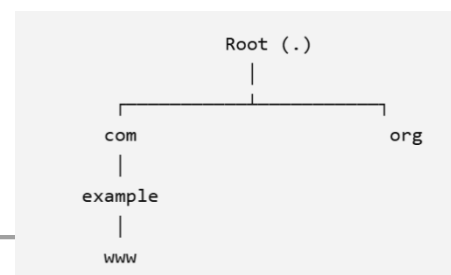
A domain name has different levels:



2. Domain Name Space

Domain name space is the hierarchical structure used to organize domain names.

The hierarchy starts from the **root**, followed by TLD, domain and host names.



3. Name Server

A **name server** is a server that stores DNS information and helps convert domain names into IP addresses.

Main types include:

- **Root Name Server**
- **TLD Name Server**
- **Authoritative Name Server**
- **Local/Recursive DNS Server**

4. Resource Record Format

A **Resource Record (RR)** contains information about a domain name.

General format:

Name | Type | Class | TTL | Data

Important record types:

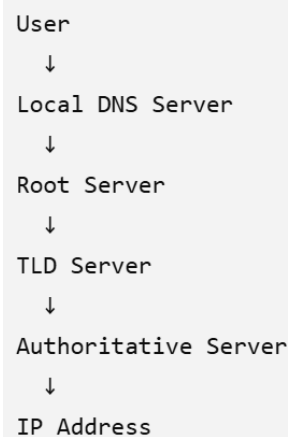
- **A:** Maps domain name to IPv4 address.
- **AAAA:** Maps domain name to IPv6 address.
- **MX:** Specifies mail server.
- **CNAME:** Gives an alias for a domain name.
- **NS:** Specifies the name server.

DNS Resolution Process

DNS resolution is the process of finding the IP address corresponding to a domain name.

Steps:

1. User enters a domain name, such as www.example.com.
2. The request goes to the **local/recursive DNS server**.
3. If the answer is not available in its cache, it contacts the **Root Server**.
4. Root server gives the address of the appropriate **TLD server**.
5. TLD server points to the **authoritative name server**.
6. Authoritative server provides the required IP address.
7. The IP address is returned to the user's computer.



Iterative and Recursive DNS Query

1. Iterative Query

In an **iterative query**, the DNS server gives the best information it has, such as the address of another DNS server. The client/recursive resolver then contacts that server itself.

Client → Root → TLD → Authoritative

Easy meaning:

👉 "You tell me where to ask next."

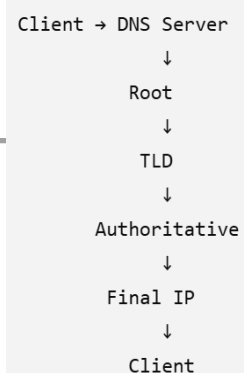
2. Recursive Query

In a **recursive query**, the DNS server takes responsibility for finding the final answer.

Client → DNS Server

Easy meaning:

👉 "You find the complete answer for me."



❖ Application Layer — 8 Marks

Definition

The **Application Layer** is the **topmost layer of the TCP/IP model**. It provides network services directly to the **users and applications**.

It acts as an interface between the **user application and the network**.

Functions of Application Layer

1. **File Transfer:** Allows users to transfer files between computers.
2. **E-mail:** Provides services for sending and receiving e-mails.
3. **Web Services:** Allows users to access websites and web pages.
4. **Remote Login:** Allows a user to access another computer over a network.
5. **Name Service:** Converts domain names into IP addresses.
6. **Resource Sharing:** Allows users to access network resources.
7. **Network Communication:** Provides communication between different applications.
8. **User Services:** Provides various network services required by users.

Services of Application Layer

Service	Purpose
Web browsing	Access websites and web pages
E-mail	Send and receive e-mails
File transfer	Transfer files between computers
Remote login	Access another computer remotely
Name service	Convert domain name into IP address

Architecture of Application Layer

The Application Layer works with the lower layers to provide network communication.

Application Layer Protocols

1. HTTP – HyperText Transfer Protocol

Used for **accessing web pages**.

2. FTP – File Transfer Protocol

Used for **transferring files** between computers.

3. SMTP – Simple Mail Transfer Protocol

Used for **sending e-mails**.

4. POP3 – Post Office Protocol version 3

Used for **receiving/downloading e-mails**.

5. IMAP – Internet Message Access Protocol

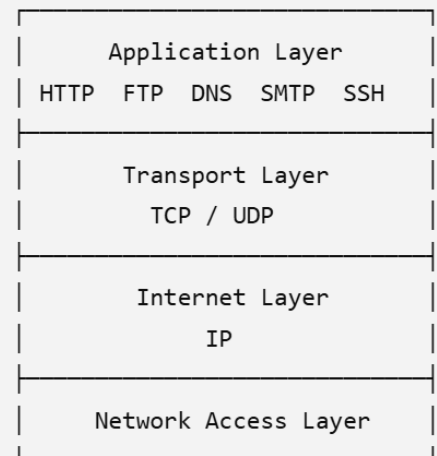
Used for **accessing and managing e-mails** from a mail server.

6. DNS – Domain Name System

Converts a **domain name into an IP address**.

7. SSH – Secure Shell

Used for **secure remote login** to another computer.



❖ WWW (World Wide Web) Architecture — 8 Marks

Definition

The **World Wide Web (WWW)** is a system of interconnected web pages and resources available through the Internet. Users access these resources using a **web browser**.

The main components of WWW are **Web Browser, Web Server, HTTP/HTTPS, URL and Web Pages**.

Components of WWW

1. Web Browser

A **web browser** is software used to access and display web pages.

Examples: Chrome, Edge, Firefox.

2. Web Server

A **web server** stores and provides web pages and other resources to users when requested.

3. HTTP/HTTPS

HTTP (HyperText Transfer Protocol) is used for communication between the browser and web server.

HTTPS provides secure communication.

4. URL

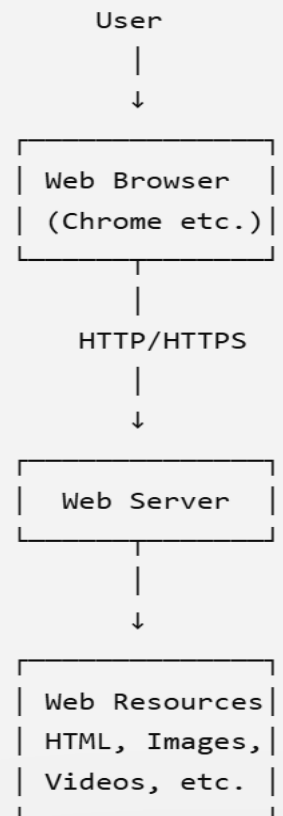
URL (Uniform Resource Locator) is the address of a web resource.

Example:

<https://www.example.com>

5. Web Page

A web page is a document displayed in a browser. It is commonly created using **HTML (HyperText Markup Language)**.



Working of WWW

1. User enters a **URL** in the browser.
2. Browser finds the IP address of the web server using **DNS**.
3. Browser sends an **HTTP/HTTPS request** to the server.
4. The web server processes the request.
5. Server sends the requested web page back to the browser.
6. Browser displays the web page to the user.

❖ Role of HTTP in WWW — 8 Marks

HTTP

HTTP stands for HyperText Transfer Protocol. It is an **application-layer protocol** used for communication between a **web browser (client)** and a **web server**.

Role of HTTP in WWW

1. HTTP allows a client to **request web pages** from a web server.
2. It allows the server to **send web pages and other resources** to the client.
3. It follows a **request-response model**.
4. It is used to transfer **HTML pages, images, videos, files**, etc.
5. **HTTPS** is the secure version of HTTP.
6. HTTP uses **TCP** for reliable communication in traditional HTTP/1.1 and HTTP/2; HTTP/3 uses QUIC.
7. It supports methods such as **GET, POST, PUT and DELETE**.
8. It uses **status codes** to indicate the result of a request, such as 200 (success) and 404 (not found).

HTTP Request Message

An **HTTP request** is sent by the client/browser to the web server to ask for a resource.

Format, Example:

Main Parts

- **Request Line:** Contains method, URL/path and HTTP version.
- **Headers:** Provide additional information about the request.
- **Body:** Contains data sent to the server, mainly with methods such as POST.

Format

```
Request Line
Headers
Blank Line
Message Body (optional)
```

Example

```
GET /index.html HTTP/1.1
Host: www.example.com
Accept: text/html
```

HTTP Response Message

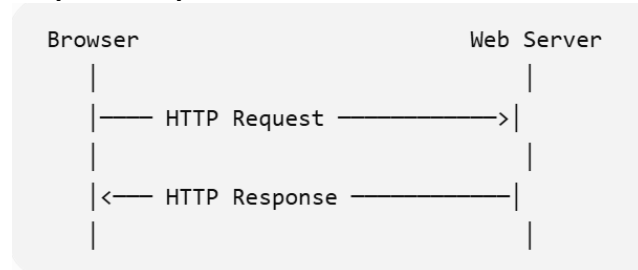
An **HTTP response** is sent by the server back to the client after receiving a request.

Format, Example:

Main Parts

- **Status Line:** Contains HTTP version, status code and status message.
- **Headers:** Give information about the response.
- **Body:** Contains the requested content such as an HTML page or image.

Request-Response Flow



Format

Status Line
Headers
Blank Line
Message Body

Example

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 500

<html>Welcome to the website</html>
```

❖ Role of SMTP in Email Transmission

SMTP stands for Simple Mail Transfer Protocol. It is an **application-layer protocol** used mainly for **sending e-mails** from a sender to a mail server and between mail servers.

Role of SMTP

1. SMTP sends an e-mail from the **sender's mail client to the mail server**.
2. It transfers e-mails between **mail servers**.
3. It uses the recipient's e-mail address to determine where the message should be delivered.
4. It uses **DNS MX records** to find the recipient's mail server.
5. It provides commands for sending the message and identifying sender and receiver.
6. SMTP commonly uses **TCP** for reliable delivery.
7. The recipient usually retrieves mail using **POP3 or IMAP**.
8. SMTP supports sending **text and attachments** with the help of MIME.

❖ Common FTP Commands and Their Functions — 3 Marks

FTP (File Transfer Protocol) is used to transfer files between a client and a server.

FTP Command	Function
USER	Specifies the username.
PASS	Provides the password for login.
LIST	Displays the list of files and directories.
PWD	Shows the current working directory.
CWD	Changes the current directory.
CDUP	Moves to the parent directory.
RETR	Downloads a file from the server.
STOR	Uploads a file to the server.
DELE	Deletes a file from the server.
QUIT	Ends the FTP session.